SNEH DUGGAL

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U.S. CITIZEN — ELIGIBLE TO WORK IN U.S. AND CANADA WITHOUT SPONSORSHIP

EDUCATION

Bachelor of Science – Computer Science | University of Victoria | BC, Canada Expected Graduation: May 2027

Coursework emphasis: Data Structures and Algorithms, Machine Learning, Software Development Methods

EXPERIENCE

AI Research Assistant (Co-op) | National Research Council of Canada | Victoria, BC Jan 2026 – April 2026

- Built a two-stage self-supervised search pipeline that indexed and made 100K+ Hubble Space Telescope images retrievable by shape/morphology, filtered from a 1M+ image archive, by engineering object-level fingerprinting and embedding retrieval
- Achieved 90%+ cluster purity on astronomical object retrieval by training VICReg embeddings and DEC clustering on 400+ class object histograms extracted via Source Extractor
- Deployed a production web demo enabling low-latency, morphology-based retrieval over the indexed archive, backed by a structured storage layer, cutting manual triage time for researchers
- Co-authored "A Self-Supervised Framework for Scalable Content-Based Search in Astronomical Data Archives," submitted to American Astronomical Society (AAS) Journals
- Presented final results and a live pipeline demo to the entire research organization, opening the search tool for astronomers to use in their own research

Undergraduate Research Assistant | University of Victoria (Remote) May 2025 – Aug 2025

- Reduced inference cost on ImageNet classification by prototyping a hierarchical routing system that dispatched "easy" queries to a quantized CLIP model and only escalated uncertain cases to the full-size model, validated via confidence-threshold sweeps
- Extended the routing approach to medical imaging by adapting it for Google's MedGemma multimodal model, running feasibility tests on MIMIC-CXR and IU-Xray datasets to measure early-stage accuracy tradeoffs
- Built swappable preprocessing, inference, and evaluation pieces so the pipeline could work on new imaging domains without rewriting the core code

Machine Learning Research Assistant | SOLIDS Lab | University of Victoria (Remote) Jan 2025 – October 2025

- Detected spoofed maritime traffic by benchmarking GNN architectures (GAE, DGI) for unsupervised anomaly detection on communication network graphs
- Built a parsing pipeline that converted raw NMEA logs into time-segmented graph structures via regex extraction, turning unstructured sensor logs into a queryable storage format for downstream modeling
- First-authored a research manuscript on GNN-based anomaly detection for spoofed maritime traffic, translating benchmark results into a reproducible methodology writeup

PROJECTS

UVic AI Course Assistant | TypeScript, Next.js, PostgreSQL, Claude API, Claude Desktop, Claude Code, Git In progress

- Built a tool-calling course-planning assistant answering real student prerequisite, eligibility, and requirement questions by orchestrating Claude over 6 deterministic tools querying a PostgreSQL schema of UVic's course and program data
- Populated the schema at scale (5,300+ courses, 560+ programs) by writing scrapers against UVic's Banner and Kuali systems, removing manual data entry as a bottleneck
- Cut multi-step response latency ~4x (45s → 12s) by parallelizing independent tool calls and adding prompt caching across the tool-use loop
- Hardened the system through adversarial testing, fixing a token-budget bug causing silent empty responses and enforcing scope boundaries against prompt-injection and jailbreak attempts
- Initiated outreach to UVic's IT Project Management team to evaluate the project and explore a path toward institutional adoption

AI Interview Coach | Python, LlamaIndex, Llama, Chroma, RAG, SpeechRecognition, Pyttsx3, Jupyter, Git

- Built a full-stack AI mock-interview coach with a FastAPI + LangChain RAG backend and an animated React/TypeScript frontend, grounding STAR-method feedback in a Chroma-embedded interview-coaching knowledge base
- Added real-time voice interaction by integrating browser speech recognition for spoken answers with OpenAI's gpt-4o-mini-tts for coach responses, driving a custom avatar that lip-syncs to live audio amplitude via the Web Audio API
- Deployed a public live demo across 3 interview tracks by provisioning a two-service architecture to Render through Infrastructure-as-Code (render.yaml) and the Render MCP server

TECHNICAL SKILLS

Languages: Python, Java, SQL (PostgreSQL)

Frameworks/Libraries: TensorFlow, PyTorch, PyTorch Geometric, OpenCV, FAISS, LlamaIndex, Chroma

AI/LLM Tools: Claude API, Claude Code, Cursor

Cloud/Tools: Git, PostgreSQL, Jupyter Notebook, Visual Studio, Microsoft Azure, AWS, Linux

Techniques: Vector Search, Vector Databases, Embedding-Based Retrieval